

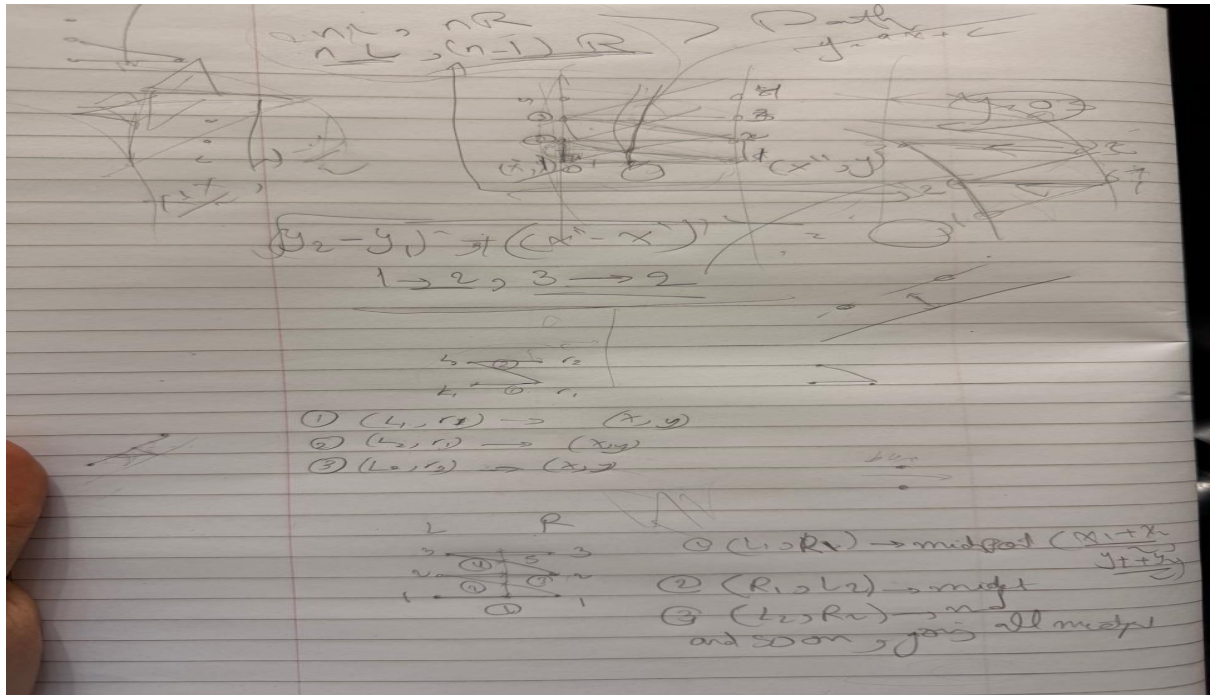
# Path Planning report

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Name: John Maged Naiem

ID:2501293

## 1) The algorithm idea used



The idea is a simpler version of delaungy triangulation where each cone on each side is connected to the one on the opposite side and the midpoint is found and then connected together as they are the path  
if no cones are on the other side the cones left are connected creating the shape of the track boundary then this line is shifted by the track half width to the middle of the track  
if no cones at all the car will continue driving forward with no chaning direction

## PART A both colours are avabile:

First blue and yellow cones are divides in separte lists

First we find the track forward direction using the fact that blue cones on left yellow on right

Sort the cones along the track direction

Choose pairs from each side and find the midpoint of the closest pair

ex: B1 Y1 ,B1 Y2 ,B2 Y2 and so one making the zigzag shape then these midpoints are connected in a rough line, divided to include many points per step not from midpoint to midpoint and check if some points is too close to a cone below a threshold then it is pushed by a certain distance ex: remaining distance to reach half track width and smoothed by taking average of before and after point making sharp corners curves and this would be the final path shape

## **PART B only one colour is available:**

- 1) Sort them by the closest to the car as no track direction can be found here
- 2) Connect them in a straight lines then find the direction of then by connecting each one to the next cone
- 3) Shift the line created by half track width to the middle using the colour of the cone to know which direction is the middle of the track
- 4) Then the same is repeated as part A the path is divided and checked if to close to cones and smoothed
- 5) A special case is only one cone is found then no line between cones is made. this cone is position is shifted to the middle of the track using its colour and this point would be the only goal to reach by the path and the track direction is assumed to be same as the car heading direction and connected with the car position like the rest of the cases and this line connecting them is then divided and smoothed

## **Limitaions:**

- 1) Miscolored cones will make this method fail
- 2) Sorting methods can fail in one sided cases especially in the hairpins or U turns
- 3) In one cone only scenario the track direction is assumed
- 4) Checking is points are close is done before the smoothing which still can make the path closer to cones but if ordered changed it might cause sudden angle changes to get away from the cone

## **Assumptions:**

- 1) track width is constant and known before in one sided cases
- 2) In one cone case the track direction is same as car heading
- 3) No miscoloured cones happen (yellow but actually blue)
- 4) Track after the scenario is a straight line

## **Note:**

8 additional scenarios are added